

POKHARA UNIVERSITY
**GANDAKI COLLEGE OF ENGINEERING AND
SCIENCE**

LABORATORY REPORT



AGILE SOFTWARE DEVELOPMENT LAB

Lab No. 4

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NAME OF EXPERIMENT

Implementation and Exploration of Agile Project Management Tools (Jira, Asana, Trello, ClickUp, and Zoho Projects).

AIM

To study and implement the basic features of Agile Project Management tools for planning, organizing, assigning, tracking, and managing software development projects.

THEORY

Agile Project Management is an iterative and incremental approach to managing software development projects. Unlike traditional waterfall models, Agile emphasizes flexibility, continuous collaboration, customer feedback, and the delivery of working software in small, frequent increments (sprints). Agile teams are self-organizing and cross-functional, and they rely heavily on transparency and communication to succeed.

To effectively implement Agile methodologies such as Scrum and Kanban, teams use specialized project management tools that help organize work, track progress, and facilitate collaboration. These tools provide a centralized platform where all project-related activities—from planning to delivery—can be managed efficiently.

The following Agile Project Management tools were explored in this experiment:

1. **Jira (Atlassian):** Jira is one of the most widely used Agile project management tools in the software industry. Developed by Atlassian, it is specifically designed for issue tracking and Agile project management. Jira supports both Scrum and Kanban frameworks, allowing teams to create user stories, manage backlogs, plan sprints, assign tasks, and generate detailed reports such as burndown charts and velocity charts. Its powerful customization and integration capabilities make it suitable for teams of all sizes.
2. **Asana:** Asana is a cloud-based project management tool that focuses on team collaboration and task organization. It allows users to create projects, break them down into tasks and subtasks, assign responsibilities, set due dates, and attach relevant files. Asana provides multiple views including lists, boards, calendars, and timelines, making it adaptable to different team preferences. Its intuitive interface and robust notification system help keep everyone aligned and accountable.
3. **Trello:** Trello is a lightweight, visual project management tool based on the Kanban methodology. It uses a board-and-card system where tasks are represented as cards

that move across lists (e.g., "To Do," "In Progress," "Done"). Trello is extremely user-friendly and suitable for small teams or individual project tracking. It supports attachments, checklists, labels, due dates, and power-ups (integrations) that extend its functionality.

4. **ClickUp:** ClickUp is an all-in-one productivity and project management platform that aims to replace multiple tools with a single integrated solution. It offers extensive features including task management, sprint planning, goal setting, time tracking, document collaboration, automation, and customizable dashboards. ClickUp supports both Scrum and Kanban methodologies and is highly configurable to fit diverse team workflows.
5. **Zoho Projects:** Zoho Projects is an online project management tool that helps teams plan projects, assign work, track issues, manage documents, and monitor project timelines. It provides Gantt charts, task boards, timesheets, and collaborative discussion forums. Zoho Projects integrates seamlessly with other Zoho applications and is often preferred by organizations already using the Zoho ecosystem.

These tools collectively address the core needs of Agile teams: transparency (everyone can see what others are working on), accountability (tasks are clearly assigned), adaptability (work can be reprioritized easily), and continuous improvement (through retrospectives and reporting).

OBJECTIVES

- To understand the core concepts and principles of Agile Project Management.
- To explore and compare different Agile Project Management tools including Jira, Asana, Trello, ClickUp, and Zoho Projects.
- To create and manage projects, tasks, and user stories using these tools.
- To assign work to team members and set appropriate priorities and deadlines.
- To track project progress using Kanban boards, Scrum boards, and dashboards.
- To understand sprint planning, backlog management, and progress monitoring.

ALGORITHM

1. Select an Agile Project Management tool from the available options (Jira, Asana, Trello, ClickUp, or Zoho Projects).
2. Create a new account or log in to the chosen platform.
3. Create a new project and provide relevant details such as project name, description, and methodology (Scrum or Kanban).
4. Define the project backlog by creating tasks, user stories, or work items.
5. Organize tasks into logical groupings using epics, milestones, or categories.
6. Assign each task to specific team members and set priority levels (High, Medium, Low).
7. Set due dates and estimated effort for each task to facilitate planning.
8. Organize tasks on the board by moving them into appropriate columns (e.g., To Do, In Progress, Review, Done).
9. If using Scrum, create a sprint and add selected backlog items to the sprint.
10. Update task statuses regularly as work progresses and team members complete their assignments.
11. Use dashboards, reports, and burndown charts to monitor overall project health and progress.
12. Perform retrospectives or reviews using the tool's collaboration features.
13. Document observations and compare features across different tools.

IMPLEMENTATION

The following activities were performed using an Agile Project Management tool (Trello and Jira were primarily used for hands-on practice, while others were studied conceptually):

1. **Project Creation:** A new software development project named "Agile Task Manager" was created on the platform.
2. **Backlog Creation:** Multiple user stories and tasks were added to the project backlog. Each item included a title, description, and acceptance criteria.

3. **Task Assignment:** Tasks were assigned to different team members based on their roles and expertise.
4. **Prioritization:** Each task was assigned a priority level to ensure that critical work was addressed first.
5. **Sprint Planning:** A two-week sprint was planned, and selected backlog items were moved into the sprint.
6. **Board Management:** Using the Kanban board, tasks were moved from "To Do" to "In Progress," then to "Review," and finally to "Done" as work progressed.
7. **Progress Tracking:** The dashboard was used to monitor task completion rates, track team workload, and identify bottlenecks.
8. **Collaboration:** Team members added comments, attached files, and updated task statuses in real-time to maintain transparency.

OUTPUT

The Agile Project Management tool was successfully used to manage project activities. The following key outcomes were observed:

- Project was created successfully with all required configurations.
- Tasks and user stories were added to the backlog effectively.
- Team members were assigned to tasks with appropriate priority levels.
- Task progress was tracked transparently using the Kanban/Scrum board.
- Project status was monitored through real-time dashboards and progress reports.
- Collaboration and communication among team members improved significantly.
- The tools helped in identifying bottlenecks and reprioritizing work efficiently.

RESULT

The implementation of Agile Project Management tools was completed successfully. The tools effectively helped in organizing project tasks, improving team collaboration, monitoring project progress, and managing software development activities in an efficient and structured manner. The hands-on experience with Trello and Jira, along with the conceptual study of Asana, ClickUp, and Zoho Projects, provided a comprehensive understanding of the available Agile tooling landscape.

CONCLUSION

This experiment provided practical and theoretical knowledge of Agile Project Management tools including Jira, Asana, Trello, ClickUp, and Zoho Projects. It is evident that these tools are indispensable for modern Agile software development teams. They simplify project planning, enhance task management, foster team collaboration, and enable real-time progress tracking. By adopting the right tool for their specific needs, teams can significantly improve their productivity, transparency, and ability to deliver high-quality software in a timely manner. Understanding these tools is essential for any software professional working in an Agile environment.